

Score-Bundle Models

Bayesian Score-Informed Performance Transcription
with Calibrated Uncertainty **[TODO: finalise title]**

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Abstract

Given a printed score and a recording of a performance (as piano MIDI in this work), we recover *how* each written note was played — its micro-timing, articulation, and dynamics — together with a calibrated statement of confidence. The core idea is a *score graph*: notes that are related in the score are modelled as related in performance, via a Gaussian Markov random field (GMRF) prior over per-note expressive variables. A small, from-scratch symbolic music language model supplies a learned prior mean, and the graph models the residual with structured, calibrated uncertainty. On held-out ASAP piano performances the full model recovers hidden notes with lower error than any per-note predictor (RMSE ≈ 0.393) and better-calibrated confidence (nominal-90% intervals cover the truth $\approx 92\%$ of the time), with the graph’s contribution statistically significant on both axes. Six downstream tasks map exactly where the structure helps (per-note recovery, confidence, error spotting) and where it does not (whole-performance summaries). **[TODO: tighten abstract; insert final numbers and the one headline sentence.]**

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Chapter 1

Introduction

1.1 Motivation

[**TODO: Why expressive performance matters; the gap between a score (what to play) and a performance (how it was played); why uncertainty is essential.**] A score specifies *what* to play; a performance realises *how*. The difference — micro-timing, articulation, dynamics — is where musical expression lives, and recovering it from a recording is both scientifically interesting and practically useful (analysis, pedagogy, expressive rendering).

1.2 Problem statement

We study *score-informed* performance transcription: the symbolic score (note identities, onsets, durations) is *known*, and the task is to infer the continuous performance variables attached to each note, with uncertainty. This is the inverse of expressive synthesis. Inferring the score itself (full automatic music transcription) is out of scope; see §1.4. [**TODO: formal one-paragraph statement; forward/inverse framing.**]

1.3 Contributions

1. A **score-graph structured prior** (GMRF/SPDE) over per-note expressive variables, giving a closed-form posterior with calibrated per-note uncertainty, validated on held-out data.
2. An **honest characterisation of a learned prior mean**: a from-scratch symbolic music LM contributes calibration- and loudness-specific information; against a rich hand-built score-feature baseline it *ties on average error* and the two *stack*.
3. A **map of where the structure helps**, via six downstream tasks with objective metrics and honest baselines (three wins, two mixed, one negative).
4. **Methodological rigour**: a discovered-and-fixed evaluation leak, a strict leakage-free protocol, and a robustness guard for the hyperparameter fit. [**TODO: phrase as a contribution, not an apology.**]
5. (*Planned / future*) A path to audio via a differentiable synthesizer likelihood (Phase 3) and a geometric “connection-Laplacian” generalisation of the graph prior.

1.4 Scope and non-goals

In scope: piano, symbolic (MIDI) performance variables (timing, articulation, dynamics), known score. Not (yet) in scope: audio input/output, intonation and vibrato (require continuously

pitched instruments), and unknown-score inference. **[TODO: expand; state assumptions explicitly.]**

1.5 Thesis outline

[TODO: one line per chapter.]

Chapter 2

Background and Related Work

2.1 Computational models of expressive performance

[TODO: survey; position our per-note, uncertainty-first framing.] [9, 7]

2.2 Score-to-performance alignment

[TODO: how aligned corpora are built; ASAP; note-level alignment.] [10]

2.3 Structured priors: GMRFs and the SPDE approach

[TODO: graph Laplacian priors; the SPDE/GMRF link; Matérn GPs on graphs.] [1, 2, 3]

2.4 Symbolic music language models

[TODO: Music Transformer, REMI, performance RNNs; from-scratch (nanoGPT) rationale.] [5, 6, 7, 8]

2.5 Differentiable synthesis

[TODO: DDSP; harmonic-plus-noise; why it matters for Phase 3.] [4]

Chapter 3

Model and Formulation

The complete derivation is in `docs/formulation.tex` (Appendix A); the essentials are reproduced here.

3.1 Score and performance variables

The known score is $\mathcal{S} = \{s_i\}_{i=1}^N$ with $s_i = (p_i, b_i, d_i)$ (pitch degree, beat onset, beat duration). Each note carries a Phase-1 performance vector

$$y_i = [\tau_i, \log r_i, v_i] \in \mathbb{R}^3,$$

the onset residual (s), the articulation/duration ratio (log), and the (normalised) velocity. **[TODO: define τ relative to the estimated tempo warp.]**

3.2 The score graph

Build a weighted graph $G = (V, E)$ over notes with edge weights from score-time, pitch, voice, and phrase adjacency, with Laplacian $L_G = D - W$. **[TODO: final edge definition; which adjacencies were used.]**

3.3 Graph prior (GMRF)

Place a Gaussian prior with precision built from the graph,

$$Q_G = \lambda I + \eta L_G \quad (\text{or Matérn/SPDE } Q_G = \sigma_g^{-2}(\kappa^2 I + L_G)^\alpha),$$

so that connected notes have correlated expressive deviations, with a closed-form Gaussian posterior. **[TODO: state per-channel construction.]**

3.4 Observation and posterior

With $\tilde{y} = y + e$, $e \sim \mathcal{N}(0, \Sigma_e)$, the posterior is Gaussian,

$$p(y \mid \tilde{y}) = \mathcal{N}(m, \Sigma_y), \quad \Sigma_y = (Q_G + \Sigma_e^{-1})^{-1}, \quad m = \Sigma_y(Q_G \mu + \Sigma_e^{-1} \tilde{y}).$$

Held-out notes carry no likelihood term and are predicted from their neighbours. **[TODO: predictive variance = latent + observation noise (the corrected form).]**

3.5 Learned prior mean and the leak-free read-out

The prior mean μ can be the LM prediction μ_{LM} : a from-scratch decoder-only Transformer, pretrained by next-token prediction on MAESTRO, read out per note. **Crucially**, the read-out is taken *before* a note’s own performed velocity enters the network, so it cannot leak the target. **[TODO: Fig.: the 4-token group and the pre-velocity read-out position.]**

3.6 Hyperparameters, fit, and robustness

Prior hyperparameters are learned by (restricted) marginal likelihood per piece. Two safeguards: a floor on the observation-noise level, and a self-validating guard that falls back to a conservative fit when the fitted field fails on a held-back sliver of observed notes. **[TODO: state both precisely.]**

Chapter 4

Data and Experimental Protocol

4.1 Pretraining corpus (MAESTRO)

[TODO: size, split, tokenisation; only MAESTRO MIDI is used for pretraining.]

4.2 Evaluation corpus (ASAP) and target extraction

ASAP is the only corpus pairing written scores with performances; it supplies the targets $y = [\tau, \log r, v]$ via the score $\leftrightarrow$ performance beat warp. [TODO: alignment/matching procedure; how many pieces/notes.]

4.3 Contamination control and the strict protocol

Because ASAP reuses many MAESTRO recordings, every ASAP performance whose MAESTRO twin was in pretraining is removed ($\approx 1000 \rightarrow \approx 650$), so the model is tested only on unseen performances. A *strict* protocol additionally blanks every hidden note’s loudness from the input. [TODO: exact counts; cost of strictness (essentially nil).]

4.4 Metrics

Recovery: RMSE (per channel and pooled). Calibration: negative log predictive density (NLL), central-interval coverage, PIT. [TODO: define; report median alongside mean (single-cell collapse guard).]

4.5 Performer-identification corpus (Vienna 4x22)

For the one task ASAP cannot support (performer identity), the small Vienna 4x22 corpus (22 pianists $\times$ 4 shared excerpts) is used. [12]

Chapter 5

Phase-1 Results

5.1 Headline: recovery and calibration

On 30 held-out ASAP pieces with 40% of notes hidden, the full model (LM guess + score graph) attains the lowest error and best-calibrated confidence; the graph’s contribution is significant on *both* axes.

5.2 The evaluation leak: discovery, fix, re-measurement

An earlier read-out let the network see a note’s own loudness — the very target it was asked to predict — inflating the apparent loudness advantage. We read the summary one position earlier (provably blind to the note’s own loudness, pinned by an automated test), re-measured everything, and adopted the strict protocol. The corrected advantage is smaller but honest and statistically significant. **[TODO: before/after loudness numbers (band-aid → leak-free → flawed claim).]**

5.3 Hyperparameter fragility and the guard

On a minority of evaluation cells the noise fit collapsed toward zero (overconfident); a noise-level floor plus a self-validating guard fix it, shown harmless on healthy runs (bit-for-bit identical). **[TODO: numbers; the 1-in-120 cell.]**

5.4 Do hand-built features suffice?

Against a rich 25-feature score baseline fit under the same protocol, the network *ties on average error*; its genuine, significant edge is calibration and loudness. Best of all, features and network *stack* (RMSE 0.388 vs. 0.393).

5.5 Stage-2: task-matched pretraining, an honest negative

A bidirectional, mask-aware retraining (matched exactly to the evaluation task) did *not* beat the original at matched compute (3× budget does not close it). The honest lesson: the original read-out already extracted what the objective change was meant to unlock. **[TODO: table; the reintroduced-leak catch.]**

Table 5.1: Held-out ASAP, strict protocol. Lower RMSE/NLL is better; coverage closer to 90% is better. [TODO: verify against docs/phase1_calibration_results.md; add NLL and CIs.]

System	RMSE	NLL	coverage
Predict zero	0.566	[TODO:]	[TODO:]
Graph alone	0.404	[TODO:]	[TODO:]
Network + graph (headline)	0.393	[TODO:]	92%
Features + network + graph	0.388	[TODO:]	[TODO:]

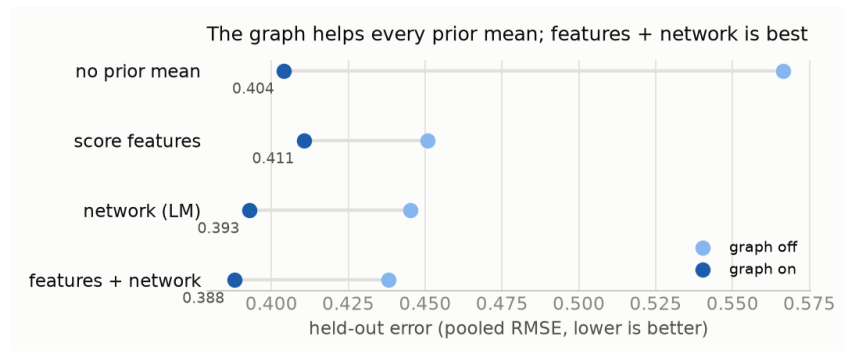


Figure 5.1: The graph’s contribution per prior mean. [TODO: final figure.]

5.6 Scaling: a bigger LM does not help

A $1.3\times$ larger, lower-perplexity model gives an essentially identical downstream result once the leak is removed — the earlier “scaling” gain was the loudness leak. Practical rule: judge the LM downstream, not by perplexity. [TODO: numbers.]

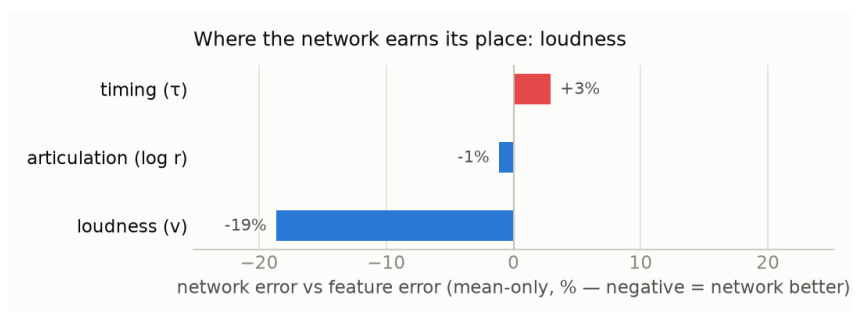


Figure 5.2: Where the network earns its place (per channel). **[TODO: final figure.]**

Chapter 6

Downstream Tasks

Six tasks with objective metrics and honest baselines; all outcomes reported. **[TODO: one results table per task; move detail from docs/downstream_tasks_results.md.]**

6.1 Error spotting (anomaly detection) — clear win

Corrupt a small fraction of notes; rank by suspiciousness. The graph model ranks better (AU-ROC **[TODO:]**) and its edge *grows* as errors get subtler.

6.2 Denoising — win at known noise level

Recover true values from added noise. With the noise level known, the graph beats per-note shrinkage on both error and calibration. *Honest negative*: estimating the noise level from a single piece is unreliable. **[TODO: numbers.]**

6.3 Completion / rendering from an excerpt — partial

Predict the expression of the rest of a piece from its opening. The LM guess carries the far-future; the graph helps only where hidden notes have nearby heard neighbours. **[TODO: numbers.]**

6.4 Selective prediction — qualified win

Discarding the least-confident predictions lowers error on the rest, most clearly for articulation and for catching the model's own misfires. **[TODO: numbers.]**

6.5 Musical-era classification — honest negative

Style statistics alone do not beat the majority-class baseline; the graph, a note-level tool, does not help whole-piece summaries. **[TODO: numbers.]**

6.6 Performer identification (Vienna 4x22) — mixed

The extracted expression *does* carry a pianist's identity (3–4× chance), validating our extraction; graph smoothing does not help this whole-performance summary. **[TODO: numbers.]**

6.7 The boundary

Across all six: **the graph prior helps whenever notes are judged one by one for accuracy or confidence, and does not help when a performance is collapsed into a summary.** This is exactly the thesis claim, now bounded by evidence on both sides.

Chapter 7

Discussion

7.1 Interpretation

[TODO: why structure helps calibration; why the LM's edge is loudness.]

7.2 Limitations

[TODO: symbolic-only; piano-only; single-corpus alignment dependence.]

7.3 Threats to validity

[TODO: leakage (addressed), contamination (addressed), fit collapse (guarded), median vs mean reporting.]

Chapter 8

Future Work

8.1 Phase 2: intonation and vibrato

[**TODO: f0 extraction from monophonic audio; new channels; datasets (URMP, singing).**]

8.2 Phase 3: a differentiable-synthesizer audio likelihood

The intended step to audio: $x = \Phi(z)\mathbf{a} + \varepsilon$, amplitudes marginalised analytically, nonlinear positions inferred approximately. [**TODO: status: not yet started.**]

8.3 Geometric direction: the score-bundle connection Laplacian

The current graph prior is the *flat* case of a graph connection Laplacian; non-trivial edge transforms give a genuine fibre bundle with measurable curvature/holonomy. [**TODO: see docs/geometry_direction.md; litmus test.**]

8.4 Unknown score support

[**TODO: toward full AMT via a discrete latent layer.**]

Chapter 9

Conclusion

[TODO: restate the validated contribution (structure + calibration, each ingredient isolated), the boundary, and the road to audio.]

Appendix A

Full formulation

[**TODO:** `\input{formulation}` here, or merge docs/`formulation.tex` into this appendix.]

Appendix B

Additional results

[TODO: per-channel tables, ablations, reliability diagrams / PIT histograms.]

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- [13] **[TODO: add remaining references (Singer–Wu connection Laplacian, Tymoczko, ATEPP, Aria-MIDI, alignment methods, ...).]**